



Module 3: Electrical Principles

Resonance · Q · Time Constants · Phasors · RF Effects
2024-2028 Amateur Extra Question Pool



Why This Module Matters

- Only 4 exam questions from E5 — but don't let that fool you
- E5 is the math foundation for:
 - → Module 5 (E7): amplifier biasing, filter design, op-amp calculations
 - → Module 7 (E9): antenna impedance, transmission line matching, Smith chart
- If you skip this math, you miss questions in those heavier modules
- What we cover:
 - → RLC resonance (series and parallel) and Q factor
 - → RC and RL time constants
 - → Phase angles in RLC circuits
 - → Rectangular and polar impedance notation
 - → Skin effect and reactive power
- Bring a calculator — we will work practice problems



Part 1: RLC Resonance and Q

Subelement E5A — 1 exam question



Series RLC Resonance

- At resonance: $X_L = X_C$ (inductive and capacitive reactances are equal)
- Series circuit behavior at resonance:
 - → Total impedance = resistance only (MINIMUM impedance — current is maximum)
 - → Voltage across L or C can EXCEED the applied voltage
 - → The Q ratio tells you how much: V_L (or V_C) = $Q \times V_{\text{applied}}$
- Resonant frequency formula:
 - $f = 1 / (2\pi \times \sqrt{L \times C})$
 - → f in Hz, L in henrys, C in farads
 - → Example: $L = 10 \text{ uH}$, $C = 100 \text{ pF}$ → $f = 1 / (2\pi \times \sqrt{10\text{e-}6 \times 100\text{e-}12}) = 5.03 \text{ MHz}$



Parallel RLC Resonance and Q Factor

- Parallel circuit at resonance:
 - → Impedance is at MAXIMUM (approximately $R \times Q^2$)
 - → Source current is at MINIMUM
 - → Circulating current between L and C is at MAXIMUM
- Q Factor:
 - → Series circuit: $Q = XL/R$ (reactance-to-resistance ratio)
 - → Parallel circuit: $Q = R/XL$
 - → Higher Q = narrower bandwidth, higher internal voltages/currents
- Half-power bandwidth: $BW = f_{\text{resonant}} / Q$
 - → Example: $f = 7.2 \text{ MHz}$, $Q = 80 \rightarrow BW = 7,200,000/80 = 90 \text{ kHz}$
 - → The -3 dB points of the resonance curve are at $\pm 45 \text{ kHz}$ from center



Part 2: Time Constants and Phase

Subelement E5B — 1 exam question



RC and RL Time Constants

- RC time constant: $\tau = R \times C$
- → After 1τ : 63.2% of final voltage charged (36.8% remaining)
- → After 5τ : effectively fully charged
- → Example: $R = 10\text{k}\Omega$, $C = 1\mu\text{F}$ → $\tau = 10,000 \times 0.000001 = 0.01 \text{ s} = 10 \text{ ms}$

- RL time constant: $\tau = L/R$
- → Same 63.2% rule applies to current rise
- → Example: $L = 10 \text{ mH}$, $R = 100\Omega$ → $\tau = 0.01/100 = 100 \mu\text{s}$

- Phase relationships — ELI the ICEman:
- → ELI: in an inductor (L), voltage (E) leads current (I)
- → ICE: in a capacitor (C), current (I) leads voltage (E)
- → Both by 90 degrees



Phase Angle in Series RLC

- Phase angle formula:
- $\tan(\theta) = (X_L - X_C) / R$
- Examples:
 - $\rightarrow X_L = 100, X_C = 60, R = 40: \tan(\theta) = 40/40 = 1.0 \rightarrow \theta = 45^\circ$ (inductive)
 - $\rightarrow X_L = 30, X_C = 80, R = 50: \tan(\theta) = -50/50 = -1.0 \rightarrow \theta = -45^\circ$ (capacitive)
 - $\rightarrow$ At resonance: $X_L = X_C \rightarrow \tan(\theta) = 0 \rightarrow \theta = 0^\circ$ (purely resistive)
- Admittance: $Y = 1/Z$ (measured in siemens)
- Susceptance: $B = 1/X$ (imaginary part of admittance)
- Conductance: $G = 1/R$ (real part of admittance)
- Polar to admittance: reciprocal of magnitude, negate the angle



Part 3: Rectangular and Polar Notation

Subelement E5C — 1 exam question



Rectangular Notation: $R + jX$

- Rectangular form: $R + jX$
- → R = resistance (real part) — plotted on horizontal axis
- → jX = reactance (imaginary part) — plotted on vertical axis
- → Positive jX = INDUCTIVE (above horizontal)
- → Negative jX = CAPACITIVE (below horizontal)

- Examples:
- → $50 + j0$: purely resistive (50-ohm load, perfect match)
- → $0 + j100$: pure inductance 100Ω
- → $0 - j100$: pure capacitance 100Ω
- → $50 + j75$: resistive + inductive
- → $50 - j25$: resistive + capacitive



Polar Notation and Conversion

- Polar form: magnitude $\angle$ angle
- $\rightarrow$ Magnitude = $\sqrt{R^2 + X^2}$
- $\rightarrow$ Angle = $\arctan(X/R)$
- $\rightarrow$ Pure inductance: $+90^\circ$ angle
- $\rightarrow$ Pure capacitance: -90° angle
- $\rightarrow$ Pure resistance: 0° angle

- Converting $Z = 30 + j40$:
- $\rightarrow$ Magnitude = $\sqrt{30^2 + 40^2} = \sqrt{900+1600} = \sqrt{2500} = 50\Omega$
- $\rightarrow$ Angle = $\arctan(40/30) = \arctan(1.33) = 53.1^\circ$
- $\rightarrow$ Polar: $50\angle 53.1^\circ$



Part 4: RF Effects

Subelement E5D — 1 exam question



Skin Effect and RF Resistance

- Skin effect: at RF, current flows near the surface of the conductor — not through the full cross-section
- → Effective resistance INCREASES with frequency
- → Skin depth decreases as frequency increases
- → At VHF/UHF: keep lead lengths short to minimize added inductive reactance
- → At microwave: keep connections short to reduce phase shift
- → Film capacitors at RF: skin effect in the foil causes loss
- Electrical length:
 - → Conductor diameter affects electrical length — larger diameter = slightly longer electrical length
- Component pitfalls at RF:
 - → Electrolytic capacitors: high parasitic inductance — don't use for RF bypass



Real vs. Reactive Power

- Real power (watts): converts electrical energy to heat (or work)
 - $\rightarrow P = I^2 \times R$ — only the resistive component dissipates power
- Reactive power (VAR — volt-amperes reactive):
 - $\rightarrow$ Energy stored in electric field (capacitor) or magnetic field (inductor)
 - $\rightarrow$ Energy is returned to the circuit — not dissipated
 - $\rightarrow$ Reactive power is 90° out of phase with real power
- In a series $R + jX$ circuit:
 - $\rightarrow$ Current is the same through all components
 - $\rightarrow$ Voltage across X is 90° out of phase with voltage across R
 - $\rightarrow P = I^2 \times R$ (NOT $I^2 \times Z$)
 - $\rightarrow$ The reactive component does not contribute to heat

Key Numbers — Module 3

Resonant frequency	$f = 1/(2\pi \cdot \text{sqrt}(LC))$
Series resonance	Minimum Z; $X_L = X_C$; max current
Parallel resonance	Maximum Z; min source current
Q (series)	X_L/R at resonance
Half-power BW	f_r / Q
RC time constant	$\tau = RC$
RL time constant	$\tau = L/R$
1 time constant	63.2% of final value
ELI the ICeman	Inductor: V leads I; Capacitor: I leads V
Phase angle	$\tan(\theta) = (X_L - X_C)/R$
Rectangular +jX	Inductive
Rectangular -jX	Capacitive
Polar magnitude	$\text{sqrt}(R^2 + X^2)$
Admittance	$Y = 1/Z$ (siemens)
Skin effect	RF resistance increases with frequency



Checkpoint Quiz 1 / 10

What is the resonant frequency of a circuit with $L = 4 \mu\text{H}$ and $C = 25 \text{ pF}$?

- A. 31.8 MHz
- B. 15.9 MHz
- C. 7.96 MHz
- D. 3.98 MHz

Answer: B — $f = 1/(2\pi\sqrt{4\text{e-}6 \times 25\text{e-}12}) = 1/(2\pi \times 1\text{e-}8) = 15.9 \text{ MHz}$



Checkpoint Quiz 2 / 10

In a series RLC circuit at resonance, what is the impedance?

- A. Zero
- B. The inductive reactance
- C. The resistance only
- D. Sum of all reactances

Answer: C — At resonance, $X_L = X_C$ and cancel; impedance = resistance only



Checkpoint Quiz 3 / 10

What is the half-power bandwidth of a circuit where $f = 3.5 \text{ MHz}$ and $Q = 70$?

- A. 245 kHz
- B. 100 kHz
- C. 50 kHz
- D. 25 kHz

Answer: C — $BW = f/Q = 3,500,000/70 = 50,000 \text{ Hz} = 50 \text{ kHz}$



Checkpoint Quiz 4 / 10

What is the time constant of $R = 47\text{k}\Omega$ and $C = 0.1\text{ }\mu\text{F}$?

- A. 4.7 ms
- B. 47 ms
- C. 0.47 ms
- D. 470 μs

Answer: A — $\tau = R \times C = 47,000 \times 0.0000001 = 0.0047\text{ s} = 4.7\text{ ms}$



Checkpoint Quiz 5 / 10

In a capacitive circuit, current relates to voltage how?

- A. Current lags voltage 90°
- B. Current leads voltage 90°
- C. In phase
- D. Current leads by 45°

Answer: B — ICE: In a Capacitor, current (I) leads voltage (E) by 90°



Checkpoint Quiz 6 / 10

What is the rectangular form of a pure 100Ω capacitive reactance?

- A. $100 + j0$
- B. $0 + j100$
- C. $0 - j100$
- D. $j100$

Answer: C — Pure capacitive reactance: $0 - j100$ (negative j indicates capacitive)



Checkpoint Quiz 7 / 10

Series circuit: $X_L=80$, $X_C=30$, $R=50$. Phase angle?

- A. 22°
- B. 45°
- C. 90°
- D. 63°

Answer: B — $\tan(\theta) = (80-30)/50 = 50/50 = 1.0$; $\arctan(1.0) = 45^\circ$ inductive



Checkpoint Quiz 8 / 10

Why are electrolytic capacitors unsuitable for RF bypass?

- A. Too low capacitance
- B. Excessive parasitic inductance
- C. Too high voltage rating
- D. Too low ESR

Answer: B — Electrolytic capacitors have high parasitic lead inductance; they act as inductors at RF



Checkpoint Quiz 9 / 10

What causes an inductor to self-resonate?

- A. Skin effect
- B. Inter-turn capacitance
- C. Core saturation
- D. Eddy currents

Answer: B — Inter-turn capacitance between adjacent windings creates a parallel LC; above self-resonant frequency it appears capacitive



Checkpoint Quiz 10 / 10

Why is real power $P = I^2R$ rather than I^2Z ?

- A. Reactance cancels at resonance
- B. Only resistance dissipates power; reactive elements return energy
- C. Z always equals R at RF
- D. Different current flows through X and R

Answer: B — Reactive components store and return energy; only R converts energy to heat



Before Module 4

Your assignment:

- Drill HamStudy E5A-E5D to 80%+ confidence
- Practice the resonance frequency calculation — it appears directly on the exam
- Know the phase angle formula: $\tan(\theta) = (X_L - X_C)/R$

Coming up:

Module 4: Circuit Components — semiconductors, diode types, digital ICs, inductors, optical devices. Heavier on memorization than math, but schematic symbol recognition is tested.